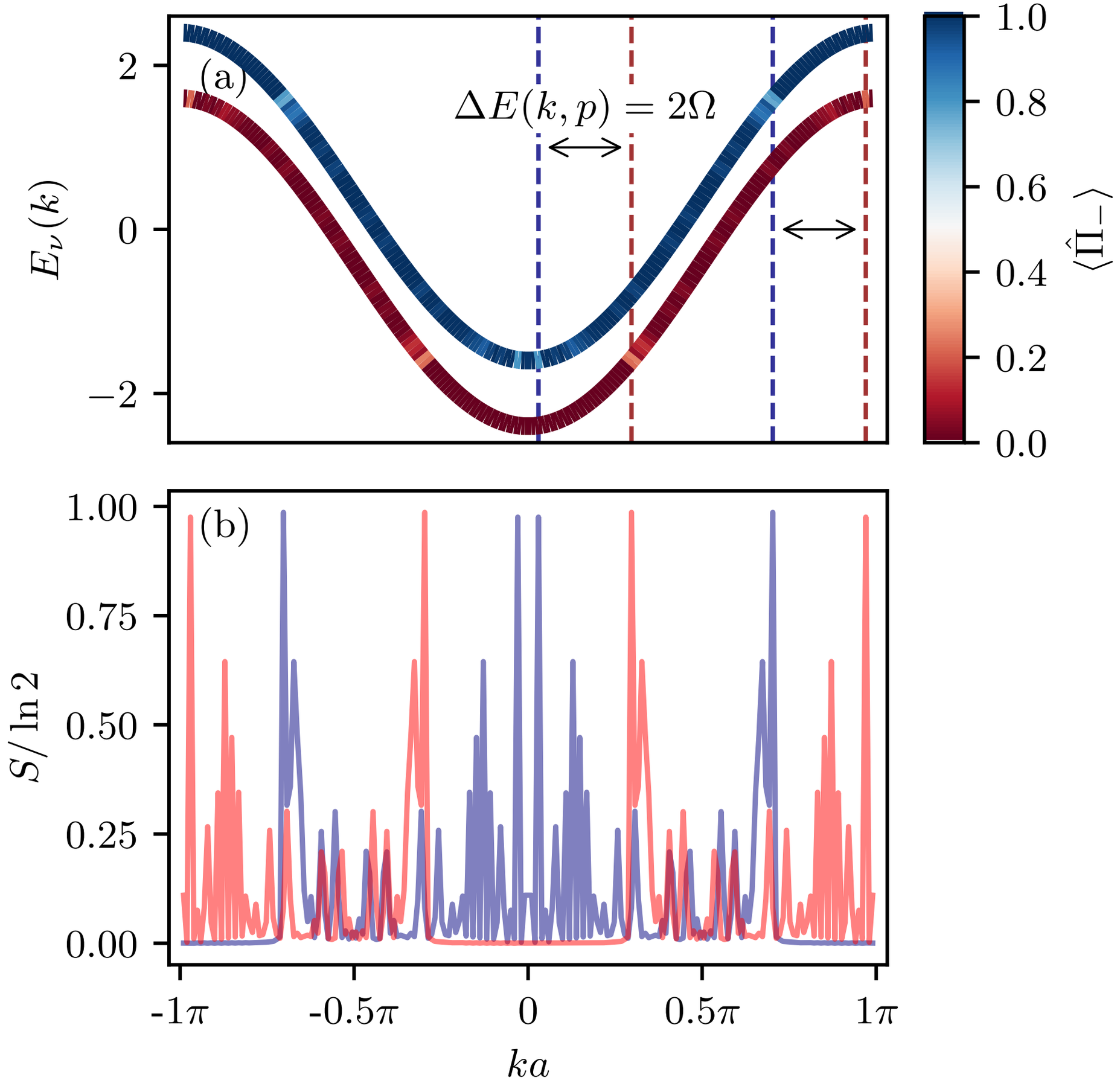

(a)
$\Delta E(k,p) = 2\Omega$
$E_\nu(k)$
2
0
−2
1.0
0.8
0.6
0.4
0.2
0.0
$\langle \hat{\Pi}_- \rangle$
(b)
$S/\ln 2$
1.00
0.75
0.50
0.25
0.00
-1π
-0.5π
0
0.5π
1π
$ka$